ELK Cluster Management and Observability Training

Duration: 5 days

A practical course for administrators and operations teams to deploy, secure, monitor and recover Elastic Stack environments. Querying supports incident investigation and operational reporting throughout the program.

Duration	Cluster management and observability	Querying
5 days / 40 instructional hours	32 hours / 80 percent	8 hours / 20 percent

Delivery: 8 instructional hours per day, excluding breaks, across five days. The 40-hour allocation includes demonstrations, guided labs, discussion and assessment. Environment provisioning is completed before Day 1.

Target audience

System administrators, SREs, DevOps engineers, platform engineers, support engineers and technical leads responsible for Elastic Stack operations.

Learning objectives

- Design a multi-node cluster and explain node roles, quorum, shards, replicas, capacity and failure domains.
- Configure authentication, TLS, least-privilege roles and Kibana Spaces; validate access boundaries.
- Build and operate ingestion pipelines using Beats, Elastic Agent, Logstash and Elasticsearch ingest pipelines.
- Apply templates, mappings, data streams, lifecycle policies and snapshot restoration to manage data safely.
- Correlate logs, infrastructure metrics, traces and synthetic checks to investigate a service incident.
- Configure monitoring rules and notification actions, and distinguish SLIs, SLOs and contractual SLAs.
- Use KQL, Query DSL and introductory ES|QL to filter, aggregate and explain operational evidence.
- Prepare sizing, upgrade, recovery and Elastic Cloud management runbooks.

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Day 5 Resilience upgrades cloud operations and capstone

Prerequisites and lab requirements

Participant prerequisites

- Basic Linux commands, processes, services, permissions and log inspection; basic Windows administration for Winlogbeat.
- Understanding of IP addressing, DNS, ports, HTTP, TLS and JSON; ability to use curl or an API client.
- Basic Docker or virtual machine familiarity; introductory monitoring and incident-management concepts.
- Basic SQL-style filtering and aggregation. Prior Elasticsearch experience is helpful but not mandatory.

Recommended lab provision per pair

Area	Training provision
Access workstation	Browser, SSH client, VS Code and API client; 8 GB RAM when using remote labs. Local labs: approximately 48 GB RAM and 250 GB free SSD.
Core cluster	Three Linux VMs, each approximately 2 vCPU, 8 GB RAM and 40 GB SSD; combined master-eligible/data roles for the lab only.
Supporting services	One Linux VM around 4 vCPU, 8 GB RAM and 60 GB SSD for Kibana, Logstash and Fleet/APM services. Stagger heavy browser-synthetic exercises.
Sources and applications	A small Linux application host with a prebuilt instrumentable HTTP service, traffic generator and logs. Shared Windows VM with permission to read event logs.
Snapshot storage	A preconfigured supported repository accessible to the required Elasticsearch nodes; shared filesystem or permitted object storage. Include restore capacity.
Network and software	Private connectivity, synchronized clocks, supported OS and a pinned compatible Elastic Stack version set; required images and packages downloaded in advance.
Security and notifications	Lab CA/certificates, role-scoped accounts, API keys, and a local webhook/mail test sink. Restrict Elasticsearch transport and API ports to the lab network.
Cloud and feature access	Instructor Elastic Cloud Hosted deployment with budget controls. Verify feature entitlements or provision a valid trial for licensed demonstrations before training.

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Trainer readiness checklist

Validate cluster formation, sample ingestion, Fleet enrollment, APM traces, synthetic browser execution, snapshot restore and the planned upgrade path before class. Prepare sanitized JSON logs, multiline traces, CSV/XML samples, Linux metrics and Windows events. Seed safe node-failure, parsing-error, latency and HTTP-error scenarios.

Resource figures are an indicative small-data training budget, not production sizing. Co-located containers do not demonstrate host-level high availability. Use staged datasets and prebuilt configurations to protect exercise time.

Day 1 Architecture deployment and security

7 hours operations and observability / 1 hour querying

Day learning objectives

- Explain how Elasticsearch, Kibana, Logstash and collectors work together; distinguish lab topology from production failure-domain design.
- Deploy and validate a secure cluster; demonstrate permitted and denied access for two roles.

Day wise schedule

Time	Track	Session including guided practice
09:00 to 10:30	Ops	ELK components, deployment choices and multi-node architecture
10:45 to 12:15	Ops	Linux deployment, discovery, node roles and health verification
13:00 to 14:30	Ops	TLS, authentication, RBAC, API keys and Kibana Spaces
14:45 to 16:15	Ops	Sizing workshop, shard allocation and licensing compatibility
16:15 to 17:15	Query	Discover, time filters, KQL and basic document retrieval
17:15 to 18:15	Ops	Node outage drill and cluster health diagnosis

Detailed topics

- Clusters, nodes, indices and documents; primary and replica shards; indexing and search paths; quorum and master eligibility.
- CPU, heap, filesystem cache, disk capacity, ingest volume, retention and shard-count inputs; benchmark before committing to a production size.

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- Kibana navigation; authentication versus authorization; Spaces versus index privileges; field-level security demonstration where entitled.
- Version history at a high level, support matrix, licensing and self-managed versus Cloud Hosted responsibilities.

Lab exercises

L1 Cluster deployment Start three pre-provisioned nodes and Kibana; capture node membership, shard placement and cluster health. Perform a controlled node outage and explain replica availability and recovery.

L2 Security boundaries Create operator and analyst roles; verify index access and Space access. Demonstrate restricted fields where licensed.

Q1 Search basics Load sample service logs and answer three questions using KQL, time filters and document inspection.

Evidence: Deliver a cluster topology, initial capacity assumptions, access-test results and saved query evidence.

Day 2 Ingestion and data lifecycle

7 hours operations and observability / 1 hour querying

Day learning objectives

- Build a reliable ingestion path and explain where parsing, enrichment and buffering belong.
- Validate a data contract and configure rollover and retention with repeatable settings.

Day wise schedule

Time	Track	Session including guided practice
09:00 to 10:30	Ops	Filebeat collection, multiple paths, multiline and processors
10:45 to 12:15	Ops	Logstash inputs, filters, outputs and pipeline validation
13:00 to 14:30	Ops	Multiple pipelines, queues, tuning and centralized management
14:45 to 16:15	Ops	Ingest pipelines, mappings, templates, data streams and ILM
16:15 to 17:15	Query	Query DSL match, term, range, bool and filter context
17:15 to 18:15	Ops	Ingestion throughput and backpressure troubleshooting lab

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Detailed topics

- Filebeat file selection and multiline boundaries; Logstash Grok, Mutate, Date and JSON exercises, with CSV/XML examples.
- Pipeline-to-pipeline communication, workers, batch size/delay, backpressure and throughput; persistent queue versus dead-letter queue scope and replay.
- Centralized Logstash management demonstration where licensed; ingest processor simulation and Logstash versus ingest-node selection.
- Static and dynamic mappings; component and index templates; data streams; ILM rollover, retention and hot/warm/cold/frozen tier concepts.

Lab exercises

L3 Reliable ingestion Parse a multiline application log, route a malformed record and inspect queue behavior under a controlled output interruption. Measure throughput and identify the limiting stage under bounded load.

L4 Data management Simulate an ingest pipeline; apply a template and an ILM policy; force a small lab rollover and inspect lifecycle status.

Q2 Query DSL Compare matches versus term and use range/bool filters to locate errors without scoring irrelevant conditions.

Evidence: Deliver pipeline configurations, a parsing-failure example, template and ILM JSON, and a validated query set.

Day 3 Infrastructure observability and investigation

6 hours operations and observability / 2 hours querying

Day learning objectives

- Collect host and stack telemetry and explain the management choices between Fleet integrations and standalone Beats.
- Correlate an ingestion or resource incident with logs and metrics, supported by reproducible queries.

Day wise schedule

Time	Track	Session including guided practice
09:00 to 10:30	Ops	Elastic Agent enrollment, Fleet policies and data streams
10:45 to 12:15	Ops	Metricbeat and Elastic Stack monitoring
13:00 to 14:30	Ops	Windows events, Linux auditing and host observability

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Time	Track	Session including guided practice
14:45 to 16:15	Query	Discover investigations and operational aggregations
16:15 to 16:45	Query	Introductory ES|QL filtering and grouped statistics
16:45 to 17:15	Ops	Monitoring dashboard review and incident evidence handover
17:15 to 18:15	Ops	Agent enrollment and monitoring gap troubleshooting lab

Detailed topics

- Fleet Server role, enrollment and agent policy lifecycle; integration assets, agent health and indices versus data streams.
- Metricbeat modules; Stack Monitoring for Elasticsearch, Kibana and Logstash; CPU, JVM, disk, queues and indexing/rejection indicators.
- Winlogbeat Windows events and Auditbeat Linux audit data; permissions and event-volume controls; avoid duplicate collection from overlapping agents.
- KQL investigation, terms/date-histogram/metric aggregations and ES|QL FROM/WHERE/STATS/SORT/LIMIT; dashboard panels and operational drilldowns.

Lab exercises

L5 Managed collection Enroll in one Elastic Agent and revise its policy; configure a Metricbeat or equivalent supported monitoring path. Diagnose a failed enrollment or monitoring gap using agent status and logs.

L6 Host incident Inspect Windows and Linux audit examples, induce bounded CPU/ingest load and identify the affected host or pipeline.

Q3 Investigation pack Build error counts by service and time, top noisy hosts and one ES|QL summary; save a concise investigation dashboard.

Evidence: Deliver an agent policy, stack-monitoring evidence, a dashboard and a query-backed incident explanation.

Day 4 APM synthetics and alerting

6 hours operations and observability / 2 hours querying

Day learning objectives

- Instrument a sample service and correlate its transactions, traces, errors and logs.

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- Configure a monitor and actionable alert, then use an SLO to describe service reliability.

Day wise schedule

Time	Track	Session including guided practice
09:00 to 10:30	Ops	APM architecture and application instrumentation
10:45 to 12:15	Ops	Heartbeat checks and synthetic monitoring locations
13:00 to 14:30	Ops	SLIs, SLOs, error budgets and rule selection
14:45 to 16:15	Ops	Connectors, rule testing, Watcher and alert operations
16:15 to 17:15	Query	Latency, errors and log to trace investigations
17:15 to 18:15	Query	Service incident query workshop and dashboard validation

Detailed topics

- APM ingestion and agent architecture, service/environment naming, sampling concepts and trace/log correlation.
- Heartbeat HTTP/TCP/ICMP concepts and lightweight monitoring; browser journeys, private locations and managed-location concepts in Synthetics.
- SLI measurement, SLO target/window and error budget; SLA as a contractual commitment rather than a generic configuration object.
- Kibana rules versus Watcher; stack-health, threshold, APM and monitor rules; action frequency, recovery actions and noise reduction; email, index and Teams connector examples.

Lab exercises

L7 APM incident Instrument the supplied service, generate latency/errors and correlate a failed request with logs and traces.

L8 Monitor and alert Create a private-location check or browser journey, trigger an outage, test a rule and record recovery. Configure an SLO where entitled.

Q4 Service evidence Filter by service/environment and calculate error rate and latency percentiles; explain the selected time window. Complete a query workshop linking latency percentiles, error-rate calculations and dashboard results to the incident timeline.

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Evidence: Deliver APM evidence, a tested monitor, an alert/recovery record and an SLO specification.

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Day 5 Recovery upgrades and capstone

6 hours operations and observability / 2 hours querying

Day learning objectives

- Demonstrate recoverability and prepare a supported upgrade plan with measurable acceptance checks.
- Resolve a combined cluster and service incident and present evidence, risks and operational handover actions.

Day wise schedule

Time	Track	Session including guided practice
09:00 to 10:30	Ops	Snapshot lifecycle, restore and controlled node-failure recovery
10:45 to 12:15	Ops	Upgrade preflight, compatibility and staged upgrade rehearsal
13:00 to 14:30	Ops	Cloud Hosted operations and capstone remediation
14:45 to 16:15	Query	Capstone investigation and operational query assessment
16:15 to 16:45	Query	Present findings with saved queries and dashboard evidence
16:45 to 17:15	Ops	Runbook review and operational assessment
17:15 to 18:15	Ops	Recovery and upgrade runbook rehearsal and review

Detailed topics

- Snapshot repositories, scheduled policies and restore validation; primary availability, unassigned shards and allocation diagnostics during node loss.
- Upgrade Assistant/deprecations, plugin/client compatibility, backups and component order for the selected release; no unsupported in-place downgrade.
- Elastic Cloud Hosted deployment, safe configuration changes, scaling, supported audit logging, upgrade controls and support-case diagnostics.
- Capstone spans ingestion, health, access, recovery and evidence-driven investigation. Serverless is compared conceptually; node administration uses self-managed labs.

Lab exercises

L9 Recovery drill Snapshot the lab index, restore it to a new name and verify data. Stop and restart one node; explain health and allocation changes.

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L10 Change and cloud drill Use a pre-staged compatible upgrade environment to rehearse checks and one upgrade step; review a Cloud Hosted change and support bundle. Rehearse the recovery and upgrade runbook with peer-reviewed acceptance checks.

Q5 Capstone Investigate an injected service issue with KQL, DSL aggregations and ES|QL; present queries plus a validated remediation/runbook.

Evidence: Deliver restore evidence, upgrade and recovery runbooks, a Cloud operations checklist and the final incident report.

Coverage and allocation

Daily hours for operations and observability versus querying: Day 1 7/1; Day 2 7/1; Day 3 6/2; Day 4 6/2; Day 5 6/2. Total: 32/8 hours, or 80/20 percent. Administrative API calls and monitoring dashboards count as operations; explicit search-language exercises count as querying. Activities are not counted twice.

Alignment to the source modules

Source module	Coverage	Day
1 to 3	ELK overview, Elasticsearch and Kibana	1
4 to 5	Heartbeat, synthetics and SLA/SLO concepts	4
6	Metricbeat and Stack Monitoring	3
7	Winlogbeat and Auditbeat	3
8	Filebeat	2
9	Logstash including pipelines, queues and tuning	2
10	Ingest pipelines	2
11	Cluster terminology and architecture	1
12	RBAC, field security and Spaces	1
13	ILM and data tiers	2
14	Mappings and component/index templates	2
15	Snapshot and restore	5

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Source module	Coverage	Day
16	Discover and visualization	1, 3, 4, 5
17	Licensing, versions and compatibility	1, 5
18 to 19	Connectors, Watcher and rule creation	4
20	APM	4
21	Elastic Agent and Fleet	3
22	Cluster sizing and architecture	1, 5
23	Cluster and component upgrades	5
24	Elastic Cloud deployment and management	5

Assessment and course summary

Assessment approach

Assess the submitted configurations, observed results and explanation of recovery decisions. Use the same 80/20 weighting as the instructional schedule. Suggested pass criterion: 70 percent overall, with successful restore validation and no unintended privileged access in the security exercise.

Assessment area	Weight
Deployment, architecture and security	20
Ingestion, lifecycle and storage operations	20
Observability, APM, monitors and alerting	25
Recovery, upgrades and cloud runbooks	15
Query correctness and investigation evidence	20
Total	100

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Course summary and participant outcomes

Participants complete an end-to-end operational workflow: deploy a secure cluster, collect and govern data, monitor platform and service health, investigate an incident, recover data and prepare a controlled change. Each participant or pair leaves with reusable configurations, saved queries, a dashboard, alert evidence and an operations runbook. Advanced production sizing, full major-version migrations and exhaustive plugin customization require follow-on practice.

Delivery and feature notes

- Daily breaks: 10:30 to 10:45 and 14:30 to 14:45; lunch: 12:15 to 13:00. Teaching runs 09:00 to 18:15. Labs are included within each scheduled block.
- Core labs are participant exercises. Advanced plugin options, licensed features, full upgrade orchestration and Cloud changes use guided demonstrations and runbook review where required.
- Verify entitlements for SLOs, Watcher, field-level security, centralized pipeline management, audit logging and external connectors. Use a licensed trainer demonstration if participant access is unavailable.
- Retain Heartbeat fundamentals for existing installations. The legacy Uptime app is deprecated; use a supported Synthetics workflow for the selected release.
- Treat Zoom integration as an approved webhook scenario, not an assumed native connector. Validate the version set and upgrade compatibility before training.

Reference material

Primary scope: ELK stack training points.docx, supplied training points, modules 1 through 24. Operational exercises receive priority; query instruction is limited to 20 percent.

Elastic subscriptions and feature availability: https://www.elastic.co/subscriptions

Elastic Uptime deprecation guidance: https://www.elastic.co/docs/solutions/observability/uptime

Elastic upgrade planning: https://www.elastic.co/docs/deploy-manage/upgrade/deployment-or-cluster

Elastic APM overview: https://www.elastic.co/docs/solutions/observability/apm

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